Assignment #3

MG-2009 Data Analysis for Business -II

Sections: BS(Fin-Tech) All Sections **Submission Date: 28/4/2026**

Total Marks:130 [125 Assignment Marks +5 Presentation Marks]

Instructions:

- Assignment should be done on A4 size paper
- There is no tolerance for plagiarism
- Write down your class section on top right corner of the title page

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Marking scheme: For all questions Hypothesis is of 1 Mark, level of significance 0.5 , Test statistics= 1.5, Critical region:2 , Conclusion:2 & calculation will be for remaining.

Q#1A manufacturing company produces the same product using two different machines and wants to compare the consistency of their output. Machine A produced a sample of 40 items with a variance in weight of 16 grams², while Machine B produced a sample of 35 items with a variance of 25 grams². Since lower variability indicates more consistent quality, management is interested in determining whether there is a significant difference in the variability of the two machines. To make this decision, the quality control team constructs a 95% confidence interval for the ratio of the two population variances and uses it to assess whether one machine is more consistent than the other.

$$\left(\frac{s_A^2}{s_B^2} \cdot \frac{1}{F_{\alpha/2,\, n_A-1,\, n_B-1}},\quad \frac{s_A^2}{s_B^2} \cdot F_{\alpha/2,\, n_B-1,\, n_A-1}\right)$$

Step 1 — Sample variance ratio:

$$\frac{s_A^2}{s_B^2} = \frac{16}{25} = 0.64$$

Step 2 — F critical values ($df_1 = 39, df_2 = 34$):

- $F_{0.025, 39, 34} \approx 2.0$ (upper critical value, used in lower bound)
- $F_{0.025, 34, 39} \approx 1.96$ (upper critical value, used in upper bound)

Step 3 — Confidence interval:

$$\text{Lower} = \frac{0.64}{2.0} = 0.32, \quad \text{Upper} = 0.64 \times 1.96 = 1.2544$$

$$\boxed{95\%\text{ CI for } \sigma_A^2/\sigma_B^2 : (0.32,\ 1.2544)}$$

The 95% confidence interval for σ²_A / σ²_B is **(0.32, 1.2544)**. Since this interval **contains 1**, we cannot conclude that the two population variances are significantly different at the 5% significance level. There is insufficient statistical evidence that one machine is more consistent than the other.

Formula:2, Steps:6, Interpretation:2 [10 marks]

Q#2: A company is evaluating the quality control of package weights produced by two different factories. Factory A has been in operation for many years, while Factory B uses more modern equipment. Management wants to determine whether there is a difference in the average weight of packages produced by the two factories. To investigate this, random samples of packages were selected from each factory, and their weights (in grams) were recorded. The sample data are as follows:

- Factory A: 118, 129, 132, 127, 113, 129, 112
- Factory B: 132, 121, 111, 123, 132, 121, 128, 113, 134, 132, 110, 108, 126

Based on this information, test the hypothesis at an appropriate level of significance to determine whether the mean weight of packages produced by Factory A is greater than that of Factory B. [15 Marks]

Step#1 Ho: $\sigma_1^2 = \sigma_2^2$

H_1: $\sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance: α=0.05

Step#3 Test Statistics

$$F = \frac{Largest\ variance}{Smallest\ Variance}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2}, largest\ variance\ df, smallest\ variance\ df}$$

$$F > F_{0.025,12,6}$$

F> 5.37

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 67.48, s_2^2 = 76.26, n_1 = 7\ , n_2 = 13, \bar{X}_1 = 122.38, \bar{X}_2 = 76.26$$

$$F = \frac{s_2^2}{s_1^2} = \frac{76.26}{67.48} = 1.13$$

1.13>5.37➔ doesn't satisfied

Step#6 Conclusion: Equal Variance

Now for mean:

Based on this information, test the hypothesis at an appropriate level of significance to determine whether the mean weight of packages produced by Factory A is greater than that of Factory B

Step#1 Ho: $\mu_1 \leq \mu_2$

$H_1: \mu_1 > \mu_2$

Step#2 Level of significance: $\alpha=0.05$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#4

Critical region

$t > t_{\alpha,n1+n2-2}$

$t > t_{0.05,7+13-2}$

$t > t_{0.05,18}$

$t > 1.734$

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 67.48, s_2^2 = 76.26, n_1 = 7\ , n_2 = 13, \bar{X}_1 = 122.38, \bar{X}_2 = 76.26$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(122.38 - 76.26) - (0)}{\sqrt{73.33\left(\frac{1}{7} + \frac{1}{13}\right)}} = 0.120$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(6 * 67.48) + (12 * 76.26)}{18} = 73.33$$

$0.120 > 1.734$ ➔ Doesn't satisfied

Step#6 Conclusion: As Cr doesn't satisfied therefore there is no evidence to reject Ho at 0.05 level of significance we conclude that there is insufficient evidence that the mean weight of Factory A packages is greater than Factory B.

Q#3: The number of grams of carbohydrates contained in 1-ounce servings of randomly selected chocolate and nonchocolate candy is listed here. Is there sufficient evidence to conclude that there is a difference between the variation in carbohydrate content for chocolate and nonchocolate

candy? Use $\alpha=0.10$.

Chocolate 29 25 17 36 41 25 32 29 38 34 24 27 29

Nonchocolate 41 41 37 29 30 38 39 10 29 55 29 [10 marks]

Step#1 Ho: $\sigma_1^2 = \sigma_2^2$

H_1: $\sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance: α=0.10

Step#3 Test Statistics

$$F = \frac{Largest\ Variance}{Smallest\ Variance}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2},v(largest),v(smallest)}$$

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 36.19, s_2^2 = 149.05, n_1 = 13\ , n_2 = 11$$

$$F = \frac{s_2^2}{s_1^2} = \frac{149.05}{36.19} = 4.11$$

$$F_{\frac{\alpha}{2},v(largest),v(smallest)} = F_{0.05,10,12} = 2.75$$

4.11>2.75➔ Satisfied

Step#6 Conclusion: As CR satisfied therefore there is evidence to reject Ho, at 0.1 level of significance we conclude that there is difference in variation.

Q#4 A financial technology company is analyzing the consistency of transaction processing times across two different digital payment systems. The company suspects that the newer system, System A, may have more variability in processing times compared to the older, more stable System B. To investigate this, random samples of transaction times were collected from both systems. A sample of 30 transactions from System A shows a variance of 8,324 (milliseconds²), while a sample of 30 transactions from System B shows a variance of 2,862 (milliseconds²). At a significance level of α=0.05, can the analyst conclude that the variation in transaction processing times for System A is greater than that for System B? Use an appropriate statistical test to justify your answer.

[10 marks]

Step#1 Ho: $\sigma_1^2 \leq \sigma_2^2$

H_1: $\sigma_1^2 > \sigma_2^2$

Step#2 Level of significance: α=0.05

Step#3 Test Statistics

$$F = \frac{Largest\ Variance}{Smallest\ Variance}$$

Step#4

Critical region

$$F > F_{\alpha,v(largest),v(smallest)}$$

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 8324, s_2^2 = 2862, n_1 = 30\ , n_2 = 30$$

$$F = \frac{s_2^2}{s_1^2} = \frac{8324}{2862} = 2.908$$

$$F_{\frac{\alpha}{2},v(largest),v(smallest)} = F_{0.05,29,29} = 1.861$$

2.908>1.861➔ Satisfied

Step#6 Conclusion: As CR satisfied therefore there is evidence to reject Ho, at 0.1 level of significance we conclude that variation in system A is larger than B.

Q#5: A fintech company is comparing the average transaction values processed through two different digital payment platforms, Platform A and Platform B. Due to differences in user behavior and system design, the variability in transaction amounts is believed to differ between the two platforms. To investigate this, independent random samples were collected. For Platform A, a sample of 20 transactions showed a mean value of $520 with a standard deviation of $80. For Platform B, a sample of 25 transactions showed a mean value of $480 with a standard deviation of $150.At a significance level of α=0.05, test whether there is a significant difference in the mean transaction values between the two platforms. [15 marks]

Step#1 Ho: $\sigma_1^2 = \sigma_2^2$

H_1: $\sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance: α=0.05

Step#3 Test Statistics

$$F = \frac{Largest\ variance}{Smallest\ Variance}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2},largest\ variance\ df,smallest\ variance\ df}$$

$$F > F_{0.025,24,19}$$

F> 2.63

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 6400, s_2^2 = 22500, n_1 = 20\ , n_2 = 25, \bar{X}_1 = 520, \bar{X}_2 = 480$$

$$F = \frac{s_2^2}{s_1^2} = \frac{22500}{6400} = 3.515$$

3.515> 2.63➔ satisfied

Step#6 Conclusion: Unequal Variance

Now for mean:

Step#1 Ho: $\mu_1=\mu_2$

$H_1: \mu_1 \neq \mu_2$

Step#2 Level of significance: α=0.05

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step#4

Critical region

$$s_1^2 = 6400, s_2^2 = 22500, n_1 = 20\ , n_2 = 25, \bar{X}_1 = 520, \bar{X}_2 = 480$$

$$df = \frac{\left(\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1}+\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} = \frac{\left(\frac{6400}{20}+\frac{22500}{25}\right)^2}{\frac{\left(\frac{6400}{20}\right)^2}{19}+\frac{\left(\frac{22500}{25}\right)^2}{24}} = 38.03$$

$t > t_{\alpha/2,df}$ or $t < -t_{\alpha/2,df}$

$t > t_{0.025,38}$ or $t < -t_{0.025,38}$

t>2.024 or t<-2.024

If critical region satisfies reject Ho

Step#5 Calculation:

$$s_1^2 = 6400, s_2^2 = 22500, n_1 = 20\ , n_2 = 25, \bar{X}_1 = 520, \bar{X}_2 = 480$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(520 - 480) - (0)}{\sqrt{\frac{6400}{20} + \frac{22500}{25}}} = 1.145$$

1.145>2.024➔ Doesn't satisfied

Step#6 Conclusion: As Cr doesn't satisfied therefore there is no evidence to reject Ho at 0.05 level of significance we conclude that the mean transaction values doesnot differ significantly between Platform A and Platform B.

Q#6: A fintech company wants to examine whether the choice of payment method is independent of customer age group. A random sample of customers was classified according to their preferred payment method (Credit Card, Mobile Wallet, Bank Transfer) and age group (Under 30, 30–50, Above 50). The observed data are summarized below:

	Under 30	30–50	Above 50
Credit Card	40	35	25
Mobile Wallet	50	30	20
Bank Transfer	30	45	35

At the 0.10 significance level, test whether payment method preference is independent of age group.

[15 marks]

$$H_o\text{: payment method preference is independent of age group}$$

$$H_1\text{: payment method preference is dependent of age group}$$

Step 2 Level of Significance : α=0.10

Step 3: Test Statistics:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where E is define as

$$E = \frac{Row\ total * Column\ total}{Grand\ total}$$

Step 4:

$$\chi^2 > \chi^2_{\alpha,(r-1)*(c-1)}$$

$$\chi^2 > \chi^2_{0.05,(3-1)*(3-1)}$$

$$\chi^2 > \chi^2_{0.10,4}$$

$$\chi^2 > 7.779$$

If Critical region satisfied reject Ho

Step 5:Calculation:

First calculate row total , Column total, then grand total, then make table

χ^2 Calculation:

Payment Method	**Under 30**	**30–50**	**Above 50**	**Row χ^2**
Credit Card	$(40-38.710)^2/38.710 =$ **0.043**	$(35-35.484)^2/35.484 =$ **0.007**	$(25-25.806)^2/25.806 =$ **0.025**	**0.075**
Mobile Wallet	$(50-38.710)^2/38.710 =$ **3.291**	$(30-35.484)^2/35.484 =$ **0.848**	$(20-25.806)^2/25.806 =$ **1.306**	**5.445**
Bank Transfer	$(30-42.581)^2/42.581 =$ **3.714**	$(45-39.032)^2/39.032 =$ **0.912**	$(35-28.387)^2/28.387 =$ **1.537**	**6.163**

Payment Method	**Under 30**	**30–50**	**Above 50**	**Row χ^2**
Col χ^2	**7.048**	**1.767**	**2.868**	**χ^2 = 11.683**

Since **11.683 > 7.779 → Reject H_0** —

Conclusion: As CR region satisfied therefore there is evidence to reject Ho, At 0.05 level of significance we conclude that payment method preference and age group are **not independent**.

Q#7: A digital lending platform is analyzing whether loan approval decisions are independent of applicants' employment status. A sample of applicants is categorized as Employed, Self-Employed, or Unemployed, along with whether their loan was Approved or Rejected. The data are given below:

	Approved	**Rejected**
Employed	120	30
Self-Employed	80	40
Unemployed	50	60

Using a 5% level of significance, test whether loan approval is independent of employment status. [15marks]

$$H_o: \text{loan approval is independent of employment status}$$

$$H_1: \text{loan approval is dependent of employment status}$$

Step 2 Level of Significance : α=0.05

Step 3: Test Statistics:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where E is define as

$$E = \frac{Row\ total * Column\ total}{Grand\ total}$$

Step 4:

$$\chi^2 > \chi^2_{\alpha,(r-1)*(c-1)}$$

$$\chi^2 > \chi^2_{0.05,(3-1)*(2-1)}$$

$$\chi^2 > \chi^2_{0.05,2}$$

$$\chi^2 > 5.991$$

If Critical region satisfied reject Ho

Step 5:Calculation:

First calculate row total , Column total, then grand total, then make table

χ^2 Calculation:

Employment status	Approved	Rejected
Employed	150×250/380 = **98.684**	150×130/380 = **51.316**
Self-Employed	120×250/380 = **78.947**	120×130/380 = **41.053**
Unemployed	110×250/380 = **72.368**	110×130/380 = **37.632**

Table 3 — Cell contributions $(O - E)^2 / E$

Employment status	Approved	Rejected	Row χ^2
Employed	$(120-98.684)^2/98.684$ = **4.607**	$(30-51.316)^2/51.316$ = **8.858**	13.465
Self-Employed	$(80-78.947)^2/78.947$ = **0.014**	$(40-41.053)^2/41.053$ = **0.027**	0.041
Unemployed	$(50-72.368)^2/72.368$ = **6.943**	$(60-37.632)^2/37.632$ = **13.342**	20.285
Column χ^2	11.564	22.227	$\chi^2 = 33.791$

$33.791 > 5.991 \Rightarrow$ satisfied

As Critical region satisfied therefore there is evidence to reject Ho, At 0.05 level of significance we conclude that loan approval is not independent of employment status

Q#8: A fintech firm is studying whether fraudulent activity is related to the type of transaction. Transactions are categorized as Online Purchase, ATM Withdrawal, or Bank Transfer, and whether they were flagged as Fraudulent or Legitimate. The observed frequencies are:

	Fraudulent	Legitimate
Online Purchase	70	230
ATM Withdrawal	40	260
Bank Transfer	30	370

At the 0.001 significance level, test whether fraud occurrence is independent of transaction type**.** [15marks]

H_o: Fraud occurrence is independent of transaction type

H_1: Fraud occurrence is dependent of transaction type

Step 2 Level of Significance : α=0.001

Step 3: Test Statistics:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where E is define as

$$E = \frac{Row\ total * Column\ total}{Grand\ total}$$

Step 4:

$$\chi^2 > \chi^2_{\alpha,(r-1)*(c-1)}$$

$$\chi^2 > \chi^2_{0.001,(3-1)*(2-1)}$$

$$\chi^2 > \chi^2_{0.001,2}$$

$$\chi^2 > 13.82$$

If Critical region satisfied reject Ho

Step 5:Calculation:

First calculate row total , Column total, then grand total, then make table

χ^2 Calculation:

(O)	(E)	(O-E)^2/E
70	42	18.67
230	258	3.04
40	42	0.10
260	258	0.02
30	56	12.07
370	344	1.97
Total		35.87

35.87>13.82

Conclusion: As CR Satisfied therefore there is evidence to reject ho at 0.001 level of significance we conclude that Fraud occurrence is dependent of transaction type.

Q#9: A company has three manufacturing plants and wants to determine whether there is a significant difference between the average ages of workers at the three locations. The following data are the ages of five randomly selected workers at each plant. Perform a one-way ANOVA to determine whether there is a significant difference between the mean ages of the workers at the three plants. Use α = 0.01 and note that the sample sizes are equal.

Worker ages		
Plant 1	**Plant 2**	**Plant 3**
29	32	25
27	33	24
30	31	24
27	34	25
28	30	26

[20 marks]

H_o: $\mu_1=\mu_2=\mu_3$

H_1: Not all 3 means are equal.

Level of Significance: α=0.01

Test Statistics:

$$F = \frac{MS_b^2}{MS_w^2}$$

Critical region: F $\geq F_{\alpha(k-1,n-k)}$ ➔ F $\geq F_{0.01(3-1,15-3)}$ ➔ F $\geq F_{0.01(2,12)}$ ➔F≥ 6.93

Calculation:

Worker ages		
Plant 1	**Plant 2**	**Plant 3**
29	32	25
27	33	24
30	31	24
27	34	25
28	30	26
141	160	124
3983	5130	3078

1) Correction Factor C.F=$\frac{(\sum X)^2}{n} = \frac{(141+160+124)^2}{15}$=12041.67
2) Total variation :$\sum X^2$ – C.F=(3986+5130+3078)-12041.67=149.33

3) Between sample :$\frac{\sum X_{.j}^2}{r}$ – C.F=$\frac{141^2+160^2+124^2}{5}$ $-$12041.67=129.73

4) Error : Total variation – Between Sample Variation=149.33-129.73= 19.60

Source of Variation	**Sum of Squares (SS)**	**df**	**Mean Square (MS)**	**F**
Between Groups (Plants)	129.73	2	64.87	39.80
Within Groups (Error)	19.60	12	1.63	
Total	149.33	14		

As CR satisfied therefore there is evidence to reject Ho at 0.05 level of significance we conclude that all plant have different average response.